

DAY 13 – PORTS & PROTOCOLS

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1. What is a network port?

A network port is a logical number used by TCP or UDP to identify a particular service or application on a device. It helps network traffic reach the correct application.

2. How many port numbers are available conceptually?

TCP and UDP each use a 16-bit port number, so each protocol has 65,536 possible port numbers, from 0 to 65,535.

3. What is a network protocol?

A network protocol is a set of rules that defines how devices communicate, including how data is formatted, transmitted, received, and interpreted.

4. What is TCP?

TCP (Transmission Control Protocol) is a connection-oriented transport protocol that provides reliable, ordered delivery of data using acknowledgements, sequencing, and retransmission.

5. What is UDP?

UDP (User Datagram Protocol) is a connectionless transport protocol with low overhead. It does not guarantee delivery, ordering, or retransmission of packets.

6. Explain two differences between TCP and UDP.

First, TCP provides reliable and ordered delivery, while UDP does not guarantee delivery or ordering. Second, TCP establishes a connection before

data transfer, while UDP can send datagrams without establishing a connection.

7. What is a listening port?

A listening port is a port on which a service is waiting for incoming network connections or datagrams. For example, an SSH service commonly listens on TCP port 22.

8. What is an ephemeral/client port?

An ephemeral port is a temporary port normally selected by the operating system for a client-side connection. It allows multiple client connections to use the same destination service.

9. What is a socket?

A socket is an endpoint for network communication. It can be described using an IP address, transport protocol such as TCP/UDP, and a port number.

10. Which port is commonly used by SSH?

TCP port 22 is commonly used by SSH (Secure Shell).

11. Which port is commonly used by HTTP?

TCP port 80 is the standard port commonly associated with HTTP.

12. Which port is commonly used by HTTPS?

TCP port 443 is the standard port commonly associated with HTTPS.

13. Which port is used by DNS?

DNS commonly uses UDP port 53 for normal queries and can also use TCP port 53, for example for larger responses or certain DNS operations.

14. Which UDP ports are commonly used by DHCP client/server?

DHCP commonly uses UDP port 67 for the server and UDP port 68 for the client.

15. What is TCP port 445 commonly associated with?

TCP port 445 is commonly associated with Microsoft SMB (Server Message Block), used for file and printer sharing and related Windows network services.

16. What is port 3389 commonly associated with?

TCP port 3389 is commonly associated with Microsoft Remote Desktop Protocol (RDP).

17. What does ss -tuln display?

ss -tuln displays listening TCP and UDP sockets. The options mean TCP, UDP, listening sockets, and numeric output respectively, so service names and hostnames are not resolved.

18. Does an open port automatically mean a vulnerability? Explain.

No. An open port only indicates that a service is reachable or listening. Security depends on the service, its configuration, software version, authentication, exposure, and whether it contains a known or exploitable weakness.

19. Why do firewalls use protocol and port information?

Firewalls use protocol and port information to identify and control network traffic. Rules can allow or block specific TCP/UDP services while permitting other required traffic.

20. Why are ports and protocols important in cybersecurity?

Ports and protocols help security professionals understand which services communicate over a network. This knowledge is useful for defensive monitoring, firewall configuration, troubleshooting, incident investigation, and authorized security assessment.

My PC has broken down, so I cannot add screenshots to some of the assignments; please do not deduct marks.